

(1) A liquid binary mixture of benzene (C_6H_6) and toluene (C_7H_8) that contains 55 wt% benzene is put into an evaporator to yield a vapor stream that is 85 wt% benzene and a residual liquid stream that is 10.6 wt% benzene.

(a) Draw a schematic of the process and label all of the streams. The mass flow rate of the feed stream is $\dot{m}_f$ and the output streams are given by $\dot{m}_v$ and $\dot{m}_l$ for the vapor and liquid streams respectively.

(b) Find the ratio of the mass flow rates of the vapor output to the liquid output ($\dot{m}_v / \dot{m}_l$).

(c) Is the problem specified, overspecified, or underspecified?

(2) Murphy 2.39

(3) Murphy 2.58

(4) Murphy 2.60

(5) Murphy 2.64

$C_6H_6 = 55\%$
 $C_7H_8 = 45\%$



$C_6H_6 = 10.6\%$
 $C_6H_6 = 55\%$

Overall Mass Balance

$$\dot{m}_f = \dot{m}_v + \dot{m}_l$$

$$0.55 \dot{m}_f = 0.85 \dot{m}_v + 0.106 \dot{m}_l$$

$$(\text{Sub } \dot{m}_f = \dot{m}_v + \dot{m}_l)$$

$$0.55(\dot{m}_v + \dot{m}_l) = 0.85 \dot{m}_v + 0.106 \dot{m}_f$$

$$0.55 \dot{m}_v + 0.55 \dot{m}_l = 0.85 \dot{m}_v + 0.106 \dot{m}_f$$

$$0.55 \dot{m}_l - 0.106 \dot{m}_l = 0.85 \dot{m}_v - 0.55 \dot{m}_v$$

$$0.444 \dot{m}_l = 0.30 \dot{m}_v$$

$$\frac{\dot{m}_v}{\dot{m}_l} = \frac{0.444}{0.30} = 1.48$$

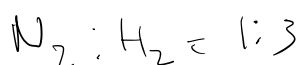
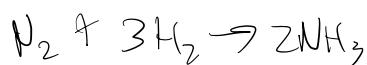
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[5/15]

2. 2.39

1440°K, 100atm

25% of N_2 reacts





$$W_2 = 1000 \times \frac{1}{2} = 500 \text{ lbmol/h}$$

$$0.25 (W_2) = 500$$

$$N_2 = \frac{500}{0.25} = 2000 \text{ lbmol/h}$$

$$H_2 = 3(2000) = 6000 \text{ lbmol/h}$$

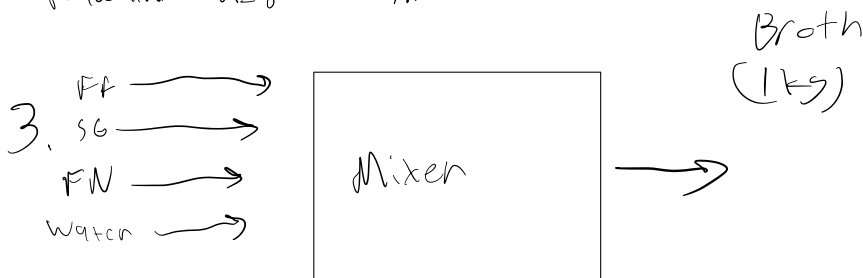
$$n = 2000 + 6000 = 8000 \text{ lbmol/h}$$

$$V = \frac{nRT}{P} = \frac{1440R \cdot 8000 \cdot 0.7302}{100} = 84119 \text{ ft}^3/\text{h}$$

$$R = 0.7302 \frac{\text{ft}^3 \cdot \text{atm}}{\text{lbmol} \cdot R} \quad T = 1440R$$

$$P = 100 \text{ atm} \quad n = 8000 \text{ lbmol/h}$$

20/20



$$FF + SG + FW + W = 1$$

Glucose Balance

$$0.45 FF = 0.15(1) = 0.333 \text{ kg}$$

Phosphate Balance

$$0.02 FF + 0.35 SG + 0.12 FW = 0.06(1)$$

$$0.02(0.333) + 0.35 SG + 0.12 FW = 0.06$$

$$0.35 SG + 0.12 FW = 0.05334$$

Nitrate Balance

$$0.01 FF + 0.15 SG + 0.58 FW = 0.6$$

$$0.01(0.333) + 0.15 SG + 0.58 FW = 0.6$$

$$0.15 SG + 0.58 FW = 0.59667$$

$$\left. \begin{aligned} 0.35 SG + 0.12 FW &= 0.05334 \\ 0.15 SG + 0.58 FW &= 0.59667 \end{aligned} \right\} \text{Two equations}$$

$$k = \frac{0.35}{0.15} = 2.333$$

$$0.15_{FW} + 0.58_{SG} = 0.05667$$

$$0.15(2.333)_{SG} = 0.35_{SG}$$

$$0.58(2.333)_{FW} = 1.353_{FW}$$

$$0.35_{SG} + 1.353_{FW} = 0.1222$$

FW

$$(0.35_{SG} + 1.353_{FW}) - (0.35_{SG} + 0.12_{FW})$$

$$= 0.1222 - 0.5333$$

$$1.2333 - 0.07887$$

$$FW = \frac{0.07887}{1.2333} = 0.06396$$

$$0.35_{SG} + 0.12(0.06396) = 0.05333$$

$$0.35_{SG} + 0.007675 = 0.05333$$

$$0.35_{SG} = 0.045658$$

$$SG = \frac{0.045658}{0.35} = 0.13045$$

$$W = 0.13045 + 0.06396 + 0.333 - 1$$

$$W = 0.472$$

$$B) (0.333 \cdot 20) + (0.13045 \cdot 10) + (0.064 \cdot 15) = \$8.92$$

We are saving
\$6.0761 per kg broth

$$6.07 \times 2000 = 121.4 \$ / \text{batch}$$

$$\text{Cost} = \$2000$$

$$\text{Batches} = \frac{2000}{121.4} = 16.48$$

17 batches
to start
seeing
real saving

$$C) m_T = 0.03_{FF} + 0.08_{SG} + 0.1_{FW}$$

$$0.03(0.333) + 0.08(0.13045)$$

$$= 0.02044 \text{ kg}$$

$$\bullet 100 = 2.04 \text{ wt\%}$$

To make 3 wt% trace nutrient
we need to increase
SUPER-GRO

30/30

$$4, \quad CO_{2w} = 5.6 \text{ Billion metric tons/yr}$$

$$US_{share} = 1/4 \text{ of } \mathcal{D}$$

$$US_{pop} = 300 \text{ million people}$$

$$US_{CO_2} = (5.6 \times 10^9)(0.25) = 1.4 \times 10^9 \text{ metric ton/yr}$$

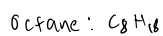
$$1.4 \times 10^9 \text{ metric tons} \times 1000 \frac{\text{kg}}{\text{ton}} \cdot 2.205 \frac{\text{lb}}{\text{kg}}$$

$$= 3.087 \times 10^{12} \text{ lb } CO_2/\text{yr}$$

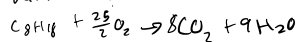
$$\frac{3.087 \times 10^{12}}{3.00 \times 10^8} = 1.029 \times 10^4 \text{ lb/person yr}$$



$$\frac{-103290}{2000} = 5.15 \text{ tons/person yr}$$



Balance Reaction



1 mol produces 8 mol CO_2

$$SC = 0.7 \text{ } \left[\begin{array}{l} \text{mole} = 1 \text{ g/mole} \end{array} \right] = 0.7 \text{ g/mL}$$

$$1 \text{ gallon} = 3.785 \text{ L} = 3785 \text{ mL}$$

$$= 2.6495 \frac{\text{kg gasoline}}{\text{gallon}}$$

Moles of gasoline

$$8(12.01) + 18(1.008) = 114.23 \text{ g/mol}$$

$$n = \frac{7649.5}{114.2} = 23.2 \text{ mol gasoline}$$

$$n_{CO} = 185.6 (44.01) = 8167 \text{ g}$$

$$= 8.167 \text{ kg } CO_2$$

$$8.167 \cdot 2.205 = 18.016 \text{ lb } CO_2$$

Given 27.5 miles/gallon average person drives

$$\frac{18.0}{27.5} = 0.655 \text{ lb } CO_2 \text{ (mile)} \rightarrow \text{person a year}$$

$$0.655 \cdot 14500 = 9497.5 \frac{\text{lb } CO_2}{\text{yr}}$$

$$10300 - 9497.5 = 802.5 \text{ lb/person yr}$$

(S/15)

5. $m_1 = \text{kg/day of } 77 \text{ wt\% } \text{H}_2\text{SO}_4$
 $m_2 = \text{kg/day of } 10.8 \text{ wt\% } \text{H}_2\text{SO}_4$
 $m_3 = \text{kg/day of pure water (0\% H}_2\text{SO}_4)$

3 unknowns

overall mass balance

$$m_1 + m_2 + m_3 = 200 \text{ kg/day}$$

H_2SO_4 balance

$$\begin{cases} 0.77m_1 + 0.108m_2 + 0m_3 = 0.18(200) \\ 0.77m_1 + 0.108m_2 = 37.2 \end{cases}$$

equation

$$\text{DOF} = 3 - 2 = 1$$

Not completely specified

77% acid: \$0.10/kg min

10.8 acid: \$0.02/kg

$$\text{Cost/day} = 0.10 F_1 + 0.02 F_2$$

concentrations stream

$$2 \frac{0.10}{0.77} = 0.1299 \text{ \$/kg H}_2\text{SO}_4$$

dilute stream

$$\frac{0.02}{0.108} = 0.1852 \text{ \$/kg H}_2\text{SO}_4$$

77% acid is cheaper

$$m_2 = 0$$

$$0.77m_1 = 37.2$$

$$m_1 = \frac{37.2}{0.77} = 48.31$$

$$m_1 + m_3 = 200$$

$$m_3 = 200 - 48.31 = 151.69$$

77 wt%

$$m_1 = 48.31 \text{ kg/day}$$

10.8 wt%

$$m_2 = 0 \text{ kg/day}$$

water

$$m_3 = 151.69 \text{ kg/day}$$

$$C = 0.10(48.31) + 0.02(0)$$

$$= 4.831 \text{ \$/day}$$

20/20

J